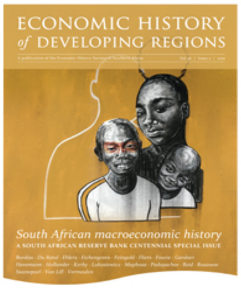


Routledge
Taylor & Francis Group

Economic History of Developing Regions

ISSN: 2078-0389 (Print) 2078-0397 (Online) Journal homepage: www.tandfonline.com/journals/rehd20

The evolution of central bank communication as experienced by the South Africa Reserve Bank

Gideon du Rand, Ruan Erasmus, Hylton Hollander, Monique Reid & Dawie van Lill

To cite this article: Gideon du Rand, Ruan Erasmus, Hylton Hollander, Monique Reid & Dawie van Lill (2021) The evolution of central bank communication as experienced by the South Africa Reserve Bank, Economic History of Developing Regions, 36:2, 282-312, DOI: 10.1080/20780389.2021.1925106

To link to this article: https://doi.org/10.1080/20780389.2021.1925106

ECONOMIC HISTORY OF DEVELOPING REGIONS
2021, VOL. 36, NO. 2, 282–312
https://doi.org/10.1080/20780389.2021.1925106

ECONOMIC HISTORY SOCIETY OF SOUTHERN AFRICA
Routledge Taylor & Francis Group
UNISA university of south africa PRESS

The evolution of central bank communication as experienced by the South Africa Reserve Bank

Gideon du Rand, Ruan Erasmus, Hylton Hollander, Monique Reid, and Dawie van Lill

Department of Economics, Stellenbosch University, Stellenbosch, South Africa

ABSTRACT
Communication has evolved into a cornerstone of central bank design and policy implementation. The South African Reserve Bank has been proactive in this regard as well – most notably with the adoption of inflation targeting in 2001. Using novel text-mining techniques, we evaluate the communication of the SARB, as presented via public speeches and monetary policy committee (MPC) statements, in the context of historical developments from 1994 to 2020. Our analysis focuses on the volume, complexity, scope, and sentiment of communication. We conclude that MPC statements are consistently and narrowly focused on the mandate, whereas speeches capture more detail about how the thinking of SARB policy makers evolves over time. In both cases, communication serves as a channel to reduce uncertainty and build credibility in the public domain.

KEYWORDS
Communication; text-mining; central bank; monetary policy; inflation targeting

JEL
E52; E58; N17; N10

Introduction

In what Blinder (2004) refers to as the 'quiet revolution', a remarkable consensus in central banking practice has emerged over the past three decades. The primary goal of monetary policy is to maintain price stability, and this goal is now predominantly achieved through an inflation targeting framework.[1] Within this framework, central bank transparency is considered essential for policy credibility and public trust (Draghi 2019; International Monetary Fund 2020).[2] Haldane, Macaulay, and McMahon (2020) describe the paradigm shift toward greater transparency as the first of two waves of the quiet revolution. This appreciation for transparency shifted central bank practice away from intentional opacity in their communication

CONTACT Hylton Hollander hollander03@gmail.com Department of Economics, Stellenbosch University, Stellenbosch 7602, South Africa

Supplemental data for this article can be accessed https://doi.org/10.1080/20780389.2021.1925106.

[1]Hammond (2012) documents 27 'fully-fledged' (official) inflation targeting central banks. As of 2021, at least 35 central banks are considered as inflation targeters – most of them official. CentralBankNews.info currently documents 73 central banks with quantitative targets or target ranges for inflation.

[2]The European Central Bank defines transparency to be the open, clear and timely provision (ie communication) of all relevant information on its strategy, assessments, policy decisions, and procedures to the general public and the markets. Academically, central bank transparency can be defined as 'the absence of asymmetric information between monetary policy makers and other economic agents' (Geraats 2002:F533).

and undue discretion in their policy decisions (Dincer and Eichengreen 2014; King 2005).[3] The second wave brought a greater focus on communication with the general public. A primary characteristic of both of these waves is the role of central bank (CB) communication as a monetary policy tool to manage inflation expectations (Blinder [2004]; Siklos [2017]; and many others).

In this paper we offer insight into the evolution of the institutional design of the South African Reserve Bank (SARB) and its policy decisions in pursuit of its mandate, as presented by the SARB itself in two of its most prominent means of communication: (1) the 135 formal statements issued by the monetary policy committee (MPC statements) from October 1999 to November 2020, and (2) 515 speeches delivered by governors, deputy governors, and other senior officials, from August 1994 to December 2020. We maintain, throughout the paper, a sensitivity to three important challenges that Reid and Siklos (2020) highlight as part of a review of the literature that focuses on the experience of emerging markets. They argue that successful communication should consider (1) the kind of message a central bank should send (quantity versus clarity); (2) recognition of the heterogeneity of the audience; and (3) how the state of the economy influences communication.

In pursuit of these research objectives, we use a range of text mining approaches to show how the *volume*, *complexity*, *scope*, and *tone* of the SARB's communication have changed over time. We measure volume of communication directly by the number of documents per year and their word counts. The literature suggests that excessive linguistic complexity undermines the clarity and accessibility of communication for a large portion of the audience, so we evaluate this feature by an often-used readability grade scoring algorithm. We analyse the scope of communication by studying the prevailing content and statistically identifying important topics via topic analysis. Finally, the tone of communication is measured by the relative proportion of positive versus negative words (a sentiment index), which provides a measure for the inclination of policy. Graham, Milligan, and Weingart (2016) argue that computer assisted textual analysis and big data offer a generative approach, in the sense that they can be used to generate new data rather than merely being used to justify a particular narrative. By using a range of these techniques, we offer different lenses through which to view the underlying information, and test the robustness of the narrative we are building.

We recognize that the meaning of words depends on context and changes through time and that there are limitations to statistical analyses that rely on word or phrase counts (Bholat et al. 2015; Shiller 2017). Furthermore, lessons of history are shown to be influential in shaping future policy responses (often used to legitimise a policy choice) as well as market reactions (Eichengreen 2012). We therefore incorporate historical events and narratives to contextualize, interpret, and validate the statistical features of our results. In doing so, we pay special attention to communication that relates to the SARB's mandate and implementation of policy, defined broadly in terms of inflation (price stability), output growth, and financial stability. Relevant historical events include changes in institutional structure (regime changes), governor tenures, international trends in the application of monetary policy, and the state of the economy.

[3]The International Monetary Fund's newly updated and revised Central Bank Transparency Code still recognizes the need for confidentiality (2020, 4). In addition to recommending a clear confidentiality policy, the IMF recommends central bank transparency be qualified for market sensitive information, financial stability considerations, and personal data.

Our main findings are as follows. The frequency of MPC statements remains very stable over time, with additional meetings only being held during periods of economic distress or to clarify institutional changes that affect the implementation of policy. In all of these cases, this communication seeks to reduce uncertainty in markets or with the public. The word count of MPC statements appears to also vary with the state of the economy (crises in particular) or with the introduction of institutional changes (such as the expansion of the mandate to include financial stability) but this is done in a prudent and predictable manner. The linguistic complexity (readability) of the MPC statements also varies over time, peaking after the exchange rate crisis and the global financial crisis, which reflects the fact that new technical concepts needed to be communicated to contextualize and explain the SARB's policy responses. Network plots then reveal that the statements are focused on inflation, growth, and expectations, which indicates a strong focus on the SARB's mandate. We are not able to identify distinctive topics with clear narrative content using topic analysis, which we ascribe to the uniform nature of the MPC statements' focus over time. But, as expected, MPC statements reflect the stance (inclination) of SARB policy more clearly than speeches. The library used for the sentiment analysis also reveals that the MPC statements use a smaller proportion of the words in the library, which again reflects their more focused nature.

In contrast, our results show that the level and variability of the volume of speeches are much higher than in the MPC statements, with the frequency of speeches increasing particularly after 2003. The word count of these documents also tends to be higher and they contain language that is generally more complex. This higher quantity and limited accessibility are very likely because these speeches are directed at more specialized audiences at universities, policy and society conferences, and the SARB's AGM, for example. We also observe that linguistic complexity of the speeches increases in the periods following the three major crises in the sample. This reflects our view that speeches were one of the tools used to clarify the SARB's role and its application of monetary policy with respect to the exchange rate, financial stability, and the Covid-19 pandemic. These speeches are likely to have been more technical in nature. Speeches also cover a more dispersed set of topics with no obvious dominant focus. Our analysis here again suggests that speeches were used during times of economic distress or to counter public pressure that could delegitimize the credibility and historical success of inflation targeting. Finally, it is difficult to interpret the sentiment index for speeches because of their variety. This broader focus is substantiated by the fact that the proportion of words matched with the dictionary is far higher.

Reflecting on these findings and their relation to historical context (academic, institutional, and economic events), we conclude that MPC statements have been narrowly focused on the mandate of the SARB and do so in a very consistent manner. There is evidence that this communication adjusts in response to crises and relevant institutional changes such as the expansion of the mandate to include financial stability, but this is done in a very prudent and predictable manner.

The speeches capture more detail about how the thinking of SARB policymakers is evolving over time. The topics addressed in these speeches are still focused on topics relevant to monetary policy, but they address more deliberately the pressures of the time. For example, we see the discussion of inflation targeting in speeches built over time, rather than dominating from the beginning. We also see that although financial stability becomes an important focus after the global financial crisis, the topic was discussed in speeches with only slightly less intensity in response to the Rand crises of 1996 and 2001. This reflects that the SARB was never blind to the importance of financial stability. Finally, the cognizance of the SARB to the need for economic growth in South Africa, and its role in providing a stable background using a flexible inflation targeting regime to enable this, is also clear.

This characterization of the SARB's communication using textual analysis provides an interesting perspective on how the SARB has presented its roles in the economy over time and the ways in which it pursues these objectives. It also allows us to explore how its institutional design and operational procedures change over time, without depending only on formal events: de jure events are presented as discrete changes, whereas in reality the de facto changes are often more gradual in nature.[4] Beyond providing a historical account of the evolution of SARB policy, this analysis is of value to a broader audience because, amongst its peers, the SARB is one of the early adopters of formal inflation targeting, making it more likely to have a good volume of text data with more deliberate communication about monetary policy.

The quiet revolution continues

Haldane (2017) captures the development of the quiet revolution (the growth of CB communication as a monetary policy tool) in two figures. The first shows that the proportion of publications about central banks that contain the phrase 'central bank communication' has increased dramatically since the early 2000s (Figure 1(a)). The second depicts the similarly steady rise in the volume of speeches by governors and other policy makers (Figure 1(b)). These developments reflect the operational role of CB communication to promote the effectiveness of monetary policy and its institutional role whereby communication is used to support institutional integrity.[5] In this section, we discuss some of the key elements of central banking theory and practice and the role of CB communication therein.[6]

In practice, central bank communication is intimately linked with the expectations channel of the monetary transmission mechanism. For example, the first experiment with inflation targeting and transparency by the Reserve Bank of New Zealand achieved remarkable success, which pioneered the design, implementation, and operation of inflation targeting (Bernanke et al. 1999). An inflation targeting regime embodies five key characteristics (see eg Hammond 2012; Mishkin 2006):

> (1) an institutional commitment to price stability as the primary, but not necessarily the only, goal of monetary policy; (2) the public announcement of a quantitative target for inflation; (3) an information-inclusive approach whereby monetary policy decisions are based on a wide

[4]Notable examples include the eventual adoption of inflation targeting in February 2000, and the announcement of the financial stability mandate in October 2010 which, seven years later, culminated in the Financial Sector Regulation Act on 21 August 2017. The FSR Act No. 9 introduced sweeping financial sector reforms and aligned South African regulatory and supervision practices with global standards.

[5]For example, it might be necessary for a governor to state a position periodically on nationalization of a central bank or monetization of fiscal debt. This form of communication protects credibility, which will obviously influence the effectiveness of monetary policy in that it supports trust, but it is directly aimed at operational objectives.

[6]For surveys on CB communication, see Blinder et al. (2008), Issing (2019), and Reid and Siklos (2020). Analysis of the reception of the CB communication from the perspective of its audiences is important in these analyses, but beyond the scope of this paper. See Reid et al. (2021).

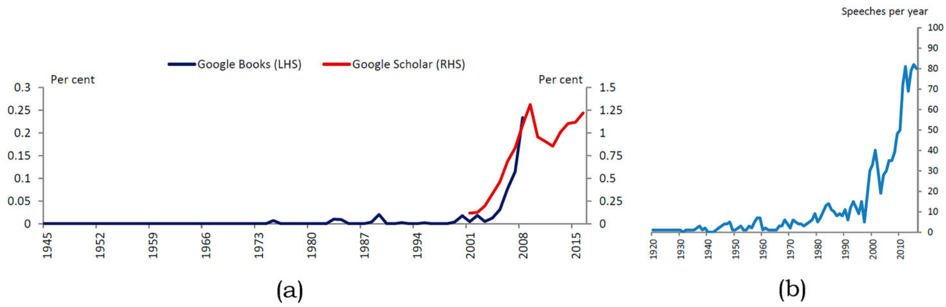


Figure 1. The 'Quiet Revolution'. (a) Publications containing 'central bank communication' as a percentage of publications containing 'central bank'. (b) Governors' and other policymakers' speeches. Source: Haldane (2017, 29 and 31).

set of information (including forecasts); (4) transparency achieved through communication with the public and the markets about the plans and objectives of monetary policymakers; (5) mechanisms of accountability.

By design, the goal-based policy framework of inflation targeting attempts to influence inflation expectations. Its characterisitcs – communication (about monetary policy), commitment (to achieve its goal), and transparency – foster the necessary credibility to anchor expectations (Bordo and Siklos 2014; Walsh 2015).

Communication, commitment, and transparency can also address the 'expectations gap' between what central banks can do and what they are expected to deliver (Bordo and Siklos 2019). Recently, the global financial crisis (GFC) led to public debate on whether it is possible, or even desirable, to maintain central bank independence (Borio 2019a; Yellen 2015), and the level of priority a central bank should give to financial stability as an objective also received considerable attention (Hollander and van Lill 2019). As a result, the question of whether a central bank can and should focus on a narrow mandate is far from over, especially for emerging market economies (Bordo and Siklos 2019; Borio 2019b). In fact, following the GFC, emerging market central banks, on average, experienced a decline in 'institutional resilience' (credibility and transparency) (Bordo and Siklos 2019).

In theory, a credible inflation target helps private sector participants (eg households, unions, firms, and financial analysts) to anchor their expectations about future inflation to the announced target. Although widely accepted to be true, there is no universally accepted theoretical framework that maps the interaction between CB communication and individuals' expectations into the monetary policy transmission mechanism (King 2005; Tarullo 2017; Blinder 2018; Reid and Siklos 2020).[7] Indeed, the GFC brought a greater emphasis on expectations management as a monetary policy tool, but Tarullo (2017) still cautions against over- relying on the expectations channel in real-time policymaking because of its underdeveloped theoretical basis. To strengthen this line of

[7]There is, of course, a well-established literature on monetary policy rules which emphasizes the importance of expectations (see eg Clarida, Galí, and Gertler 1999; Woodford 2003; Walsh 2015) and a vast empirical literature on central bank credibility and transparency which shows that institutional design aimed at improving coordination between policy makers and private sector decision makers improves the effectiveness of monetary policy (see eg Gürkaynack, Sack, and Swanson 2005a, 2005b, 2007; Larrain 2005; Reid 2009; Gürkaynak, Levin, and Swanson 2010). However, this literature is diverse, and no universal consensus has arisen.

research, a recent strand of literature explores the role of expectations formation in the conventional new-Keynesian framework that is founded on the strict assumption of full-information rational expectations (Coibion, Gorodnichenko, and Kamdar 2018; Palley 2012). The main thrust of this research agenda is to incorporate real-time expectations data in alternative models of the expectations formation process. More recently, there is a growing literature exploring the role of narratives (such as CB communication) in causing and sustaining economic fluctuations (Shiller 2017). In this literature, the increasing use of textual analysis by economists to analyse large databases of CB textual data (Bholat et al. 2015) offers additional tools to study the role of communication and expectations in monetary policymaking.

Notably, the quiet revolution occurred in spite of the underdeveloped nature of the theory about how inflation expectations are actually formed.[8] So, while the motivation for using CB communication is widely accepted, reflected in rapidly increasing volumes of CB communication, there remains a great need to better understand the formation of inflation expectations and the manner in which CB communication can be used to influence them.

For communication to be valuable (for it to influence behaviour), it must be received, understood, and trusted by the intended audience (Haldane 2017). If that is the case, we need to be sensitive to the heterogeneity of the audience (the second challenge highlighted by Reid and Siklos (2020)). Originally, CB communication was mostly directed at a relatively small group of financial experts. The second wave of the revolution increased the focus on CB communication with the general public, which elevated the concern about how central bank audiences receive its communication, if at all, and whether these audiences understand its communication (Haldane, Macaulay, and McMahon 2020). That is, clarity of communication outweighs the mere increase in volume of communication – the first challenge (Reid and Siklos 2020). While the question of whether the communication is understood involves studying the reception of the message, which is beyond the scope of this paper, there is evidence to suggest that reducing the linguistic complexity of the communication will increase its clarity (Bulíř, Cĭhák, and Jansen 2013; Siklos 2003; Winkler 2002). This should improve the chances that CB communication is received and understood by the audience of a central bank.

The use of measures of linguistic complexity, such as the Flesch-Kincaid reading grade score, in academic papers (Bulíř, Cĭhák, and Jansen 2013) as well as speeches by central bank officials (Haldane 2017) to evaluate CB communication is becoming more widespread. There is also evidence that some central banks are deliberately aiming to use this feature to improve accessibility of their communication. For example, the Bank of England began publishing a broader-interest version of its quarterly Inflation Report in 2017 as part of a layered approach, whereby they have two versions of the document aimed at different audiences (Haldane and McMahon 2018).

The use of topic analysis is becoming more widely used to explore how the content of CB communication evolves (see eg Coco and Viegi (2020) and Siklos, Amand, and

[8]There is an ongoing discussion about whether CB communication should be viewed as a substitute or complement to traditional monetary policy tools (Siklos 2017). Reid and Siklos (2020) argue that it should remain primarily a complement, despite the fact that during unusual times it may need to play a stronger role.

Wajda (2018)). This allows analysts to ask questions about the themes that were discussed at a particular time and how these themes relate to each other contemporaneously and over time. This can give some interesting insight into the narratives being crafted by a central bank.

Finally, sentiment analysis, in the context of CB communication, is typically used to capture the communication about the likely future path of policy (for studies that link SARB communication with its policy decisions see Reid and Plessis (2010) and Erasmus and Hollander (2020)). These kinds of analyses are useful in that they provide insight into how central banks map the state of the economy into their narratives to justify policy decisions and guide market expectations.

Mining the communication of the South African Reserve Bank

We use a range of text-mining techniques to characterize the communication of the SARB over the period 1994 to 2020. Our analysis focuses on the volume, readability, scope, and sentiment of CB communication over the period. Our data consists of a large set of CB communication documents from the SARB, taking the form of unstructured textual data. It includes all of the 133 monetary policy committee (MPC) statements in the period from October 1999 to July 2020; and 515 speeches by governors, deputy governors, and other officials of the South African Reserve Bank delivered between August 1994 and March 2020.[9]

Textual analysis of CB communication offers insights into the narratives that dominated at the time and how these relate to monetary policy design and implementation (Shiller 2017).

For example, policy changes can be monitored at a higher frequency, which allows for the characterization of policy changes as an evolution rather than as discrete adjustments (of the policy rate) at points in time. This distinction can be understood from Shiller's hypothesis that narratives randomly mutate like diseases studied within evolutionary biology. In addition, while it is easy to collect information about de jure economic and institutional developments that are likely to have influenced the mandate of the SARB over time, the relevant de facto changes in institutions often do not match these de jure events precisely (eg Reinhart and Rogoff 2004). For example, the implementation of a policy framework such as inflation targeting can develop naturally over time out of the need for a nominal anchor for price stability (eg the case of New Zealand (Bernanke et al. 1999)) or it can vary across countries in non-negligible ways (eg whether a quantitative target is publicly announced or not (Hammond 2012)), and the influence of political pressures on economic policy and institutions could also be relevant (Cukierman and Webb 1995; de Haan and Eijffinger 2016).

[9]Our sample of speeches includes the governor's annual address to SARB shareholders and several speeches by executive general managers (senior management). As the data was scraped from the SARB website, the speeches were recorded in different formats: pdf, text, and html. It was therefore not possible to obtain a complete data structure on the speaker, official topic, and audience. By inspection, we observe that the corpus of speeches is dominated by governors and deputy governors, and the fora in which speeches take place varies widely (eg universities, policy and society conferences, international summits, private sector functions, and press statements). We do not quantify this aspect of communication, which is left for future research. All speeches and monetary policy committee statements are made available to the public at the SARB's website: https://www.resbank.co.za/.

The systematic analysis of the content of CB communication using computational approaches is also a valuable complement to manual content analysis (ie reading and extracting meaning) (Bholat et al. 2015; Erasmus and Hollander 2020; Reid and Plessis 2010; Siklos, Amand, and Wajda 2018) and other narrative approaches (Romer and Romer 1989). First, computational approaches can analyse large collections of documents (corpora) far more rapidly than any human reader; and, because the pre-processing and computational code can be made public, the results are easier to replicate. Second, computer-enabled approaches to classifying text offer a relatively more objective approach than manual classifications, which may add valuable insights. Graham, Milligan, and Weingart (2016) argue that big data and the use of computer assisted textual analysis offer the economic historian a 'macroscope' to complement the standard practice in economic history. The macroscope begins with a large quantity of complex data and selectively reduces it until 'once obscure patterns and relationships become clear' (Graham, Milligan, and Weingart 2016, 1). This contrasts with the standard historical approach, in which the researcher begins with small evidence and extracts knowledge through expansion.

However, a fully computational approach is not a panacea. While human readers may be more subjective in their approach, they can follow systemic, rigorous approaches to coding data (Krippendorff 2013) and identify themes that are not as easily identified by computational approaches. The validity of good manual coding can be higher than that achieved through computer assisted approaches, because computer-enabled approaches lack the ability to understand context or perceive the intent of language (Shiller 2017). In addition, computational algorithms perform better with more data, so short pieces of text can be a challenge. Therefore, to support the relative strengths of the automated text-mining approach used in this paper, we use a range of techniques to test robustness and we evaluate the results with strong reference to the narratives at the time (the historical context).

In recognition of these points, we adopt an exploratory approach to analysing the evolving mandate of the SARB (as reflected through its institutional design), and we offer insights into the implementation of policy to meet this mandate. In doing so, we consider the state of the economy when analysing communication (the third challenge identified by Reid and Siklos (2020)), as well as the fact that 'the meanings of words depend on context' (Shiller 2017, 998). That is, to validate our computational findings and to assist with our interpretation thereof, we discuss our results within the context of historical events (economic and institutional) and developments in the academic literature. We also include quoted excerpts from our sample, in line with Shiller's suggestion that sometimes the exact wording of a narrative is best preserved as it is to convey the narrative of the time. Below, we provide brief descriptions of each of the four methods used to evaluate the SARB's communication.

Volume

We begin by measuring the volume of communication by the SARB from 1994 to 2020: the number of MPC statements and speeches and their respective word counts over time. Our focus here is to illustrate how the extent and type of communication has changed with the international trend towards increased transparency and CB communication. This provides a first view of the SARB's participation in the Quiet Revolution.

Readability

Text that is longer or uses more complex language is less accessible to a greater proportion of the public. Therefore, clarity of communication, and not simply an increase in its volume, is required to improve policy effectiveness. To evaluate the linguistic complexity (readability) of the language used in SARB speeches and MPC statements over time, we use the Flesch-Kincaid reading grade score (Flesch 1948; Kincaid, Braby, and Mears 1988). This measure has already been used to evaluate the clarity and accessibility of CB communication of other central banks (see Haldane (2017) and citations therein) but has not yet been applied to the communication of the SARB.

The Flesch-Kincaid score is designed to generate an index calibrated to the US school system and assesses the approximate reading grade level of a text – ie the approximate number of years of education required to easily comprehend a piece of text. For example, a score of 8.4 indicates a text that an 8th grade student should be able to read. A higher score means that a particular piece of text is more difficult to read. Notably, the test does not have an upper bound by construction, however, the 13–20+ range is typically associated with a college or graduate-level degree of difficulty. The test relies on three variables: (1) total number of words, (2) total number of sentences, and (3) total number of syllables. In essence, the test determines the complexity of both sentences and words within the text using these variables. In turn, these two factors are weighted according to pre-determined parameters, and the result is then adjusted by a constant to arrive at the grade level score.[10]

Topic Analysis

After we describe the changing volume and accessibility of SARB communication over the Quiet Revolution, we characterize the scope (content) of this communication in order to evaluate how the SARB has presented its role (mandate) within the economy. We use Latent Dirichlet Allocation which is an unsupervised machine learning algorithm that estimates a probabilistic distribution of words over topics and topics over documents (Blei, Ng, and Jordan (2003); Bholat et al. (2015); Siklos, Amand, and Wajda (2018); see Wehrheim (2019) for further detail about how the approach is being used within digital humanities and economic history). This allows the topics covered in the speeches and the prominence each received to be estimated and analysed statistically. We use an inductive approach to organize and discuss the statistical information from the machine learning methods into themes that are related to the SARB mandate in order to explore de facto changes rather than de jure ones.[11] This allows us to address questions such as how the SARB governors have portrayed the role of the SARB (for example, the relative levels of attention paid to inflation, output growth, and financial stability over time) and how they describe the implementation of monetary policy (for example, the instruments and communication strategies used).

[10]The Flesch-Kincaid grade level score is determined according to the following model: Flesch-Kincaid Grade Level = 0.39*(total number of words/total number of sentences) + 11.8*(total syllables/total number of words) − 15.9.

[11]'Induction is a third epistemology that, like abduction, starts from data without priors, but like deduction, then seeks to generate general theoretical claims' (Bholat et al. 2015:1).

Sentiment

Finally, we analyse the sentiment of the speeches using the 'bag-of-words' approach employed by Erasmus and Hollander (2020). The methodology first isolates each individual word in a document, which are then matched to a predefined sentiment library that consists of positive and negative words – akin to a positive economic outlook and a negative economic outlook. Therefore, in the context of monetary policy, more positive ('hawkish') words than negative ('dovish') words in an MPC statement indicates a greater likelihood of increasing the policy rate, and vice versa. For the purpose of this paper we use the Henry (2008) sentiment library, which is a library specifically tailored for financial applications.[12] Subsequently, the difference between the total number of hawkish and dovish matches are computed and standardized by the total number of matches to arrive at an index value that is reflective of the relative sentiment of the SARB (ie hawkish or dovish) expressed in a particular document. The index is scaled by a factor of two such that it fluctuates between 2 and −2 – the former indicating the most hawkish sentiment while the latter indicates the most dovish sentiment.[13] This process is repeated for every document in the corpus, which enables us to analyse how the relative sentiment of the SARB has evolved over time.

The historical evolution of SARB communication

The SARB, like most other central banks, employs a range of communication tools to improve the efficiency of monetary policy to achieve its policy goals and to be accountable to the public. Most studies of SARB communication have focused on the MPC statements. In this paper, we evaluate both the MPC statements and a range of speeches by the SARB governors and deputy governors. There is reason to believe that these modes of communication may differ in some respects, given the different audiences to which they are likely to be directed as well as the purpose of the interactions. For example, formal MPC statements only begin in October 1999 as part of the new monetary policy framework of inflation targeting. Including speeches therefore allows us to study SARB communication over a longer period and across institutional frameworks.

Notably, our sample period includes an important period where CB communication was evolving into a pillar of the institutions of monetary policy (International Monetary Fund 1999). Figure 2 shows that the period under review (1994–2020) begins four years after the Reserve Bank of New Zealand becomes the first central bank to formally adopt an inflation targeting framework, and six years before the SARB does so. In 1999, Bernanke et al. (1999) publish their well-known academic contribution to the case for inflation targeting. In 2004, Blinder (2004), an academic and a policymaker, writes about the 'quiet revolution' he has been witnessing, and, in 2008, Blinder et al. (2008) survey the literature and identify a focus on CB communication with the public as an area that requires more attention within academia. By 2017, Binder publishes a survey

[12]Erasmus and Hollander (2020) find this sentiment library to be superior in terms of explaining future interest rate movements.

[13]The index is scaled purely to make direct comparisons to existing literature. The sentiment index (SI) is calculated according to the following formula: $SI = 2 * (\textit{Number of Hawkish Words} - \textit{Number of Dovish Words})/(\textit{Number of Hawkish Words} + \textit{Number of Dovish Words})$.

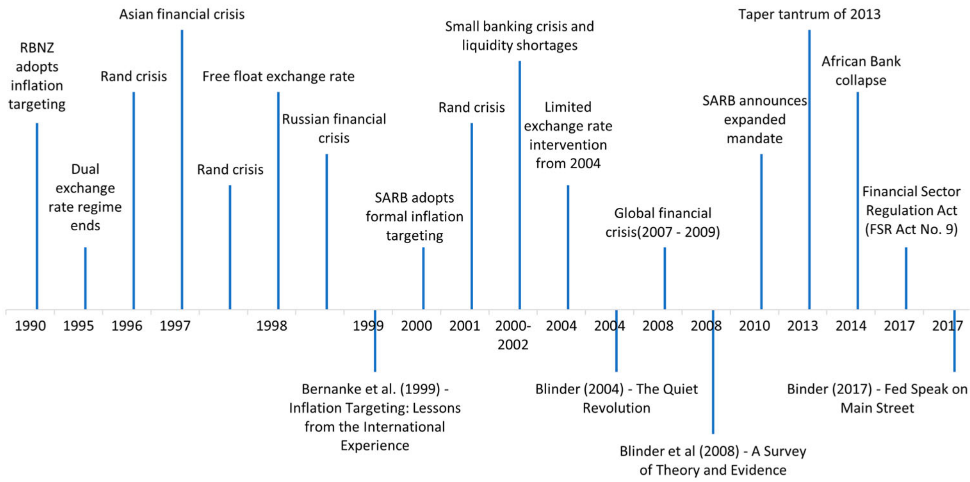


Figure 2. The historical context of SARB communication: historical events and seminal papers.

of the academic literature on CB communication with households, reflecting progress within academia on the second wave.

The sample period also includes a number of economic and institutional events such as the global financial crisis and the announcement by the SARB of an expanded mandate to include financial stability. These developments provide valuable historical context for interpreting the SARB's communication about its role in the economy.

Volume

Following the establishment of the Monetary Policy Committee (MPC) and its first meeting on 13 October 1999, the MPC 'reconfirmed the Reserve Bank's commitment to ensure the widest possible consultation process as part of implementing inflation targeting' (24 November 1999). Indeed, the goal of the SARB during this time was to build credibility to anchor inflation expectations within the mid-point of its' inflation target band of 3–6% by 2002:

> The target range of 3%–6% set was an ambitious one [(see Figure 3)]. Whilst there may be debates today about whether the target range was set at the correct level, it is important to note that for both the credibility of the central bank, as well as the management of expectations in respect of inflation being a problem, the target range had to be set at a level which would properly demonstrate commitment to lowering inflation. – Tito Mboweni, Governor of the SARB from 1999 to 2009. (Mboweni 2003)

With the formal adoption of inflation targeting in February 2000, regular MPC statements became a feature of monetary policy in South Africa, with little variation in volume over the period, apart from additional meetings during times when the economy was experiencing a shock (see Figure 4). We can however, see that the total number of speeches made by the SARB increased notably from 2003. Shortly thereafter, observed inflation fell within the target band for several years (see Figure 3).

This apparent focus on greater transparency and the management of expectations shows a sensitivity to the need to communicate more broadly than only with financial experts,

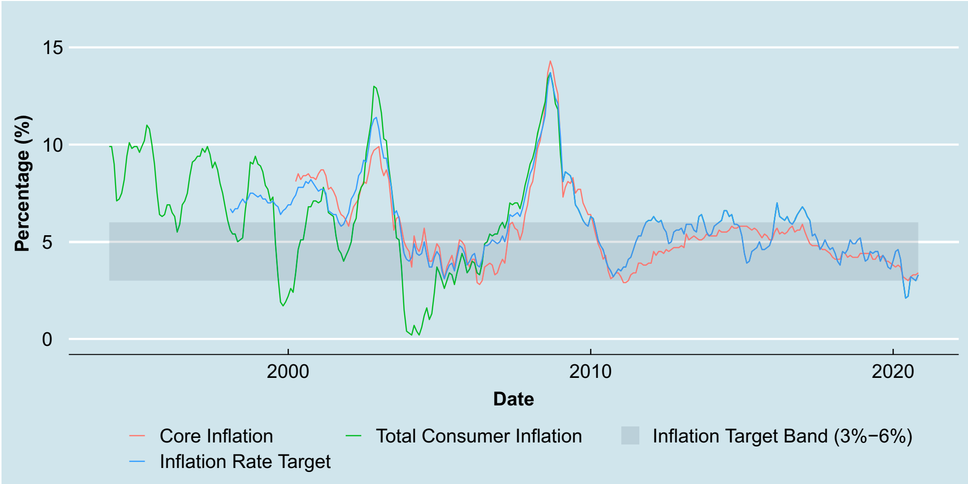


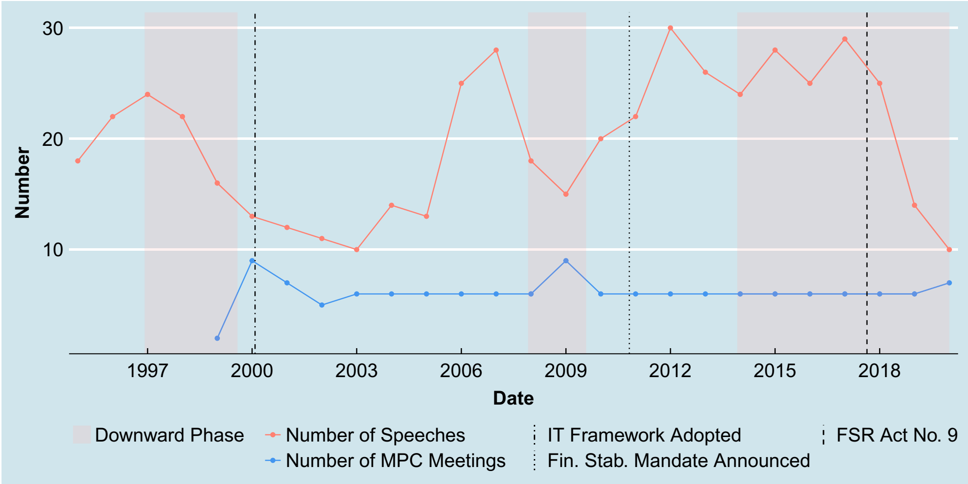
Figure 3. Historical inflation series and the inflation target band.
Note: The series for the inflation rate target is the official inflation rate that the SARB targets to be within the 3–6% band.

despite the fact that the second wave of the quiet revolution only truly begins after 2008, with the challenge by Blinder et al. (2008). For example, this monetary policy regime change coincided with other institutional changes, such as the first Monetary Policy Forum (MPF) held on 20 March 2000 in the SARB Head Office in Pretoria. Mr Tito Mboweni, then Governor of the South African Reserve Bank, outlined the intent of the MPF as follows:[14]

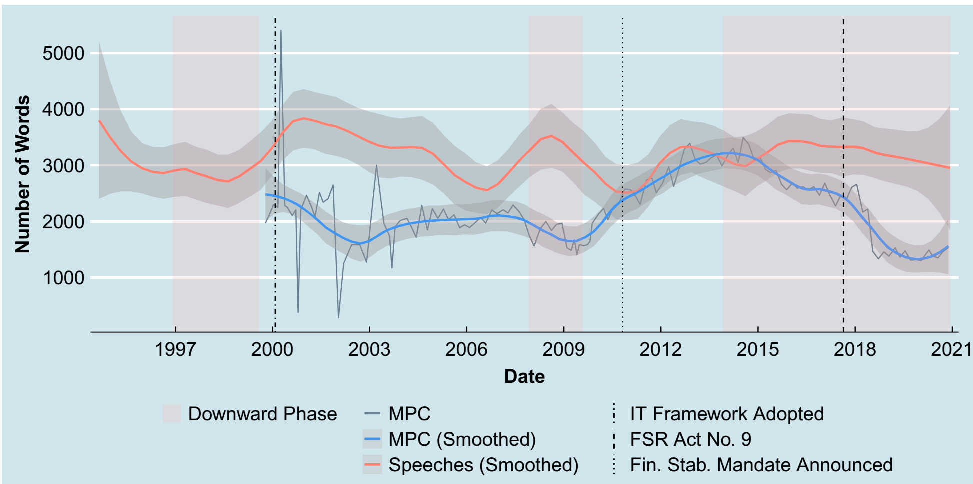
> The MPF is a mechanism for interaction and communication between the SARB and outside stakeholders who take interest in the economic and financial developments locally and internationally [Its objectives are] to enhance transparency in the decision-making process around monetary policy, to open the avenue for constant discussions on monetary policy and general economic developments here and abroad, to ensure that the views of organised entities are taken into account in the determination of monetary policy, and, to ensure a healthy, on-going debate and mutual exchange of views on macro-economic policy. In attendance were various organised entities, including business, the trade union movement, community organisations, private sector economic analysts, the chairpersons of the Finance Committee in the National Council of Provinces and the National Assembly. Also present were Reserve Bank deputy governors, senior managers and other staff members. A wide-ranging discussion on recent monetary policy developments as well as general economic developments was held. The Monetary Policy Forum will meet twice a year and more stakeholders will be invited in future.[15]

Clearly, the intention to consider the audience to whom the SARB communicates became an important facet of its CB communication agenda. That is, CB communication is not only a tool for monetary policy operating procedures, it is a component of the institutional design of a central bank.

[14]The statements and speeches we use as data source were all retrieved from the SARB website (South African Reserve Bank 2020). For brevity, we do not cite or reference them individually.
[15]And in line with the conditional commitment that central bankers often take when managing inflation expectations (Blinder 2018), the Governor noted that although [econometric] models by themselves are not enough to enhance the chances of meeting the inflation targets, they are nevertheless an important tool in ensuring that the future path of inflation is measured with a certain degree of certainty.



(a)



(b)

Figure 4. Volume of speeches and MPC statements. (a) Frequency of speeches and MPC statements. (b) Word count of MPC statements and speeches.

Notes: Eight MPC meetings were scheduled for 2000 and a special meeting was convened on 16 October. The April 2000 MPC meeting contains a detailed appendix discussion on the new inflation targeting framework. Thereafter, MPC meetings were scheduled approximately every six weeks (with the exception of 2002). Special meetings were held in January 2002 (currency crisis) and April 2020 (Covid-19). In 2009, monthly meetings were held from February to November, with the exception of July (global financial crisis). The special meeting held in January 2002 contains only a single paragraph. For presentation purposes, we only plot the smoothed series for speeches because the word count varies substantially by speech.

Our results show that the frequency of *speeches* declines during downward phases of the business cycle and major economic crises, while their word count tends to remain elevated (Figure 4).[16] This absolute decline in the volume of speeches might suggest

[16]The absolute word count varies substantially by speech, but the smoothed series suggests some relation with the state of the economy or with the introduction of institutional changes (Figure 4(b)). The volume of speeches do not appear to

greater consolidation or consensus in communication as a means of fostering a consistent message. One reason to (tentatively) draw this conclusion is that during episodes of economic distress the number of *MPC statements* increases above the usual allocation of six statements per year.[17] These additional meetings were as follows:

(1) A special meeting was convened on 16 October 2000 in response to expected higher inflation due to a strong rand depreciation and sustained high petroleum prices. Not long thereafter, the 2001 currency crisis hit, and another special meeting was held in January 2002.
(2) During the global financial crisis turmoil, monthly meetings were held in 2009 from February to November, with the exception of July.
(3) A special meeting was held in April 2020 in response to the ongoing coronavirus (Covid-19) pandemic.

Notably, the first two of these episodes speak to each of the two major institutional (mandate) changes that took place and the benefits that the they produced in the formulation of monetary policy – including CB communication. First, South Africa experienced exchange rate crises in 1998 and 2001, which manifested from both domestic and international instability. In a special meeting of the MPC on 16 October 2000, the SARB only marginally tightened their policy stance in response to second-round inflationary concerns in September 2000, due to rand exchange rate depreciation and sustained increases in petroleum prices. In the following year the SARB's non-interventionist response to the intensive pressure on the rand was remarkably different than their interventionist and unsuccessful attempt to quench the rapid exchange rate depreciation in mid-1998 (Bhundia and Ricci 2006). The sterilized intervention in 1998 quickly proved ineffective, in part, because it was hard to build confidence after the global currency crisis. The announced intentions to adopt inflation targeting turned out to be well timed. In fact, in an International Monetary Fund review of economic developments in South Africa at the time, Bhundia and Ricci (2006) applaud the SARB for their correction:

> The South African authorities have acknowledged that the intervention policy in 1998 was inappropriate. They avoided adopting the same intervention policy in 2001, which proved to be a very successful strategy as the macroeconomic repercussions of the crisis were limited and the rand strengthened over the next few years. The successful management of the currency crisis in 2001 was a reflection of a broader improvement in the overall macroeconomic policy framework, which helped to strengthen policy credibility (172).

The exchange rate crisis tested how resolute the SARB was to implement their new inflation targeting regime. They passed with distinction.

At this point, it is important to note that the SARB explicitly recognized financial stability as part of their new monetary policy framework:[18]

be influenced by governor tenures (Figure S5 in the online Appendix). That said, tenure changes have typically followed very soon after economic downswings; therefore, a more-sophisticated analysis should be done, which we leave for future research.

[17]Eight MPC meetings were originally scheduled for 2000. Thereafter, MPC meetings would be scheduled approximately every six weeks (with the exceptions of 2002, 2009, and 2020).

[18]Statement of the new monetary policy framework as explained in the online Appendix to the April 2000 MPC statement.

> The primary objective of monetary policy is to protect the value of the currency in order to obtain balanced and sustainable economic growth in the country. This objective is articulated in both the Constitution of the Republic of South Africa and in the South African Reserve Bank Act, No 90 of 1989. It requires the achievement of financial stability, ie price stability as well as stable conditions in the financial sector as a whole.

The financial crises in 1997 and 1998 provided an impetus to communicate financial stability 'as an important precondition for sustainable high growth and employment creation' and as a way for the SARB to 'make their unique contribution to general economic development in South Africa' (MPC statement, April 2002):

> If financial institutions and markets are uncertain or unstable it is difficult to produce, consume and invest, and therefore to increase employment. The recent emerging- market financial crisis in 1997 and 1998 has also clearly illustrated that foreign investment can be withdrawn easily and in large amounts from countries that investors perceive as high-risk destinations. Moreover, it is difficult for a country with a high rate of inflation to remain competitive in a global environment where more and more countries have already successfully reduced inflation to low levels.[19]

In sum, this communique was a clearly articulated move away from the self-described 'eclectic monetary policy approach in which intermediate objectives still played a prominent role'. Second, amidst the ongoing GFC turmoil, the SARB announced in March 2009 that they would hold their regular MPC meetings more often 'so as to allow for more effective planning and quicker reactions to changes in the economy'. These regular meetings highlighted the SARB's efforts to communicate their policy stance to the general public, manage inflation expectations, and contextualize their role in fostering financial stability. Thereafter, throughout 2010, the global economic outlook continued to be characterized by 'heightened uncertainty … in the wake of the European sovereign debt crisis, high rates of unemployment in the US and the euro area, and weak demand in many of the advanced economies [amongst other factors]' and risk in the domestic banking still appeared 'to be higher than was the case before the crisis' (MPC Statement, September 2010).

Within the context of sustained pressure on financial stability, the SARB emphasized their conditional commitment to their projected policy stance: 'Our approach remains forward-looking and is informed by close examination of the data and future developments' (Gill Marcus, 9 September 2010). One month thereafter, during the 27 October 2010 Medium Term Budget Policy Statement, the (then) Minister of Finance Pravin Gordhan announced an expanded mandate of the SARB. This mandate sanctioned the SARB to maintain and enhance financial stability.[20] Although South African monetary policy entered a new era, it does not appear that the SARB intended to deviate from the status quo although there was mounting pressure to do so. On 18 November 2010 Governor Gill Marcus made an unusual response to heightened public discourse/debate on the appropriate actions the SARB should take:

[19]Please note that the quotations need to be interpreted as individual datapoints, forming part of a large series of such pieces of text from the speeches and MPC statements. Conclusions were not drawn from these illustrative quotations alone.

[20]The Financial Sector Regulation Act (FSR Act No. 9) on 21 August 2017 introduced sweeping financial sector reforms and aligned South African regulatory and supervision practices with global standards (see Hollander and van Lill [2019] for a review of the SARB's financial stability policies).

> In considering the prevailing conditions and data, it is important to emphasise that the focus, notwithstanding any immediate impact or influence, is on the situation that might exist 12–18 months hence. There are significant voices and varied opinions aired in the public domain, all expressing a view on what needs to be done. While it is important to listen to and hear what is being said, it is the responsibility of the Monetary Policy Committee to determine the path of interest rates without fear or favour. And this is what we will continue to do.

Although the frequency of speeches have risen over time, and the introduction of inflation targeting initiated the regular release of MPC statements, the increased usage of this form of CB communication does not imply greater clarity.[21] In fact, a combination of greater volume and greater complexity might be counterproductive. We turn to assessing the complexity of the MPC statements and speeches in the following section.

Readability

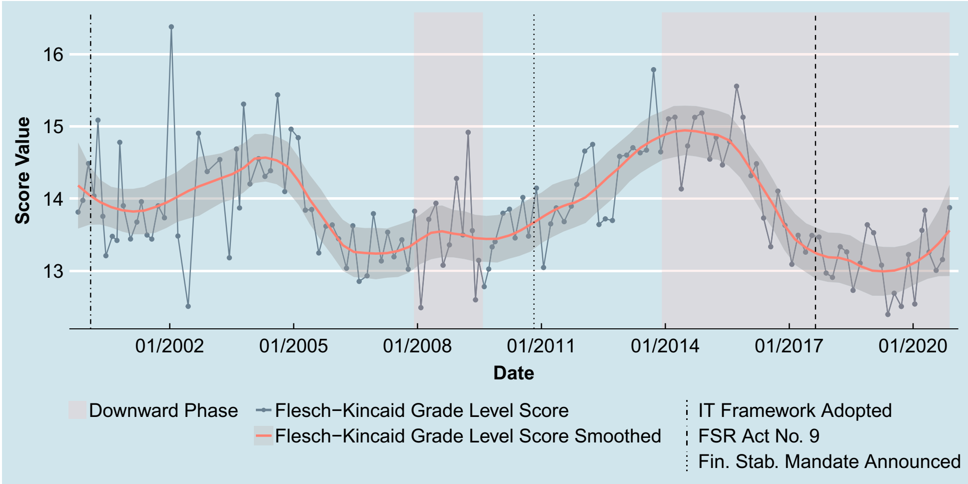
The Flesch-Kincaid grade level scores for the MPC statements and the speeches are presented in Figures 5 and 6 respectively. The document-level score is the grey line. To get an idea of the trends, we fit a local regression (LOESS).[22] First, although we observe a large degree of variability in both sets of text, speeches cover a much wider range of grade scores (note the difference in the scaling of the vertical axes). This result is intuitive given the relatively standard structure and purpose of MPC statements.[23] The trends show clearly that speeches have hovered around a grade score of 16, whereas MPC statements are constrained between a score of 13–15. Both the level and relative readability scores for MPC statements are in line with the findings across other central banks (Haldane 2017).

Besides the differences in variability between the two corpora, their trends are broadly similar, except for two periods. The speeches suggest that linguistic complexity decreased with the adoption of inflation targeting. The complexity of the speeches then steadily declined until the global financial crisis, whereas that of the MPC statements increased around the time of the exchange rate crisis. The complexity of both corpora of text increases for a number of years, beginning at the time of the global financial crisis. It seems quite reasonable that this was due to the financial instability and the adoption of the additional mandate of financial stability. The announcement of the financial stability mandate and FSR Act No 9 are represented by dotted lines in these figures. This was a substantial change to the role of central banks and there is still no consensus on the long run efficacy and implications of the new tools introduced. Finally, both corpora show a decrease in complexity beginning in about 2015, but that of the MPC statements again

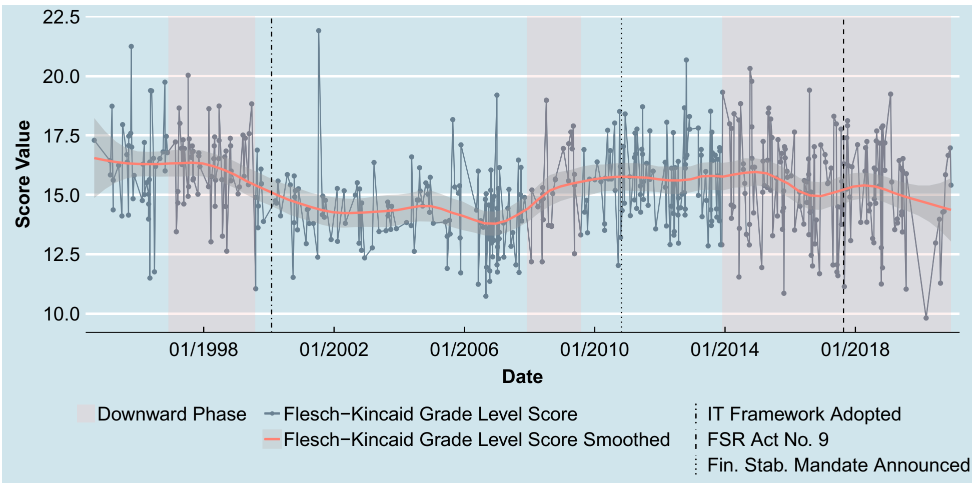
[21]We should also note that our dataset excludes other new forms of communication that begun over this period, such as monetary policy forums. In this sense, it may under-represent the increase in volume of CB communication.

[22]Locally estimated scatterplot smoothing (LOESS) is a regression approach that yields a smooth function constructed as a moving average of linear slope estimates constructed from a moving window of observations.

[23]The physical layout (structure) of MPC statements has been quite consistent since its inception. We identified four layout styles. From 1999 to 2001 there were typically six or seven sections describing domestic and international developments, the financial sector, external balance, financial stability, and monetary policy. For 2002, the layout was presented more as a narrative, with no headings, but still containing numbering of key points. From 2003 to 2010, MPC statements contained focused section headings on inflation (inflation outcome/developments and inflation outlook) and a section on the monetary policy stance. From 2011, MPC statements became more narrative in nature, with no headings, and, until recently, typically longer than previous periods (see Figure 4(b)). In March 2013, the MPC started referring to their statements as 'press statements'. These series of changes to the presentation of the statement appear to coincide with the SARB's two major institutional changes and the two waves of the quiet revolution.



(a)



(b)

Figure 5. Flesch-Kincaid (readability) grade scores for MPC statements and speeches. (a) MPC statements (with economic cycles). (b) Speeches (with economic cycles).

Notes: MPC statements cover the period from October 1999 to November 2020 and speeches cover the period from August 1994 to December 2020.

begin to rise from March 2020, very possibly in response to challenges caused by the pandemic (SARB MPC Statement, March 2020).

It is also quite common in the literature to consider how different governors affect CB communication (eg in Siklos, Amand, and Wajda 2018). Over this period in South Africa, Tito Mboweni was SARB governor from August 1999 to November 2009, Gill Marcus from November 2009 to November 2014, and Lesetja Kganyago from November 2014 to the present. It is not obvious that the tenures of these governors drives changes in readability of the SARB communication we study (see Figures S6b and S6a in the online Appendix).

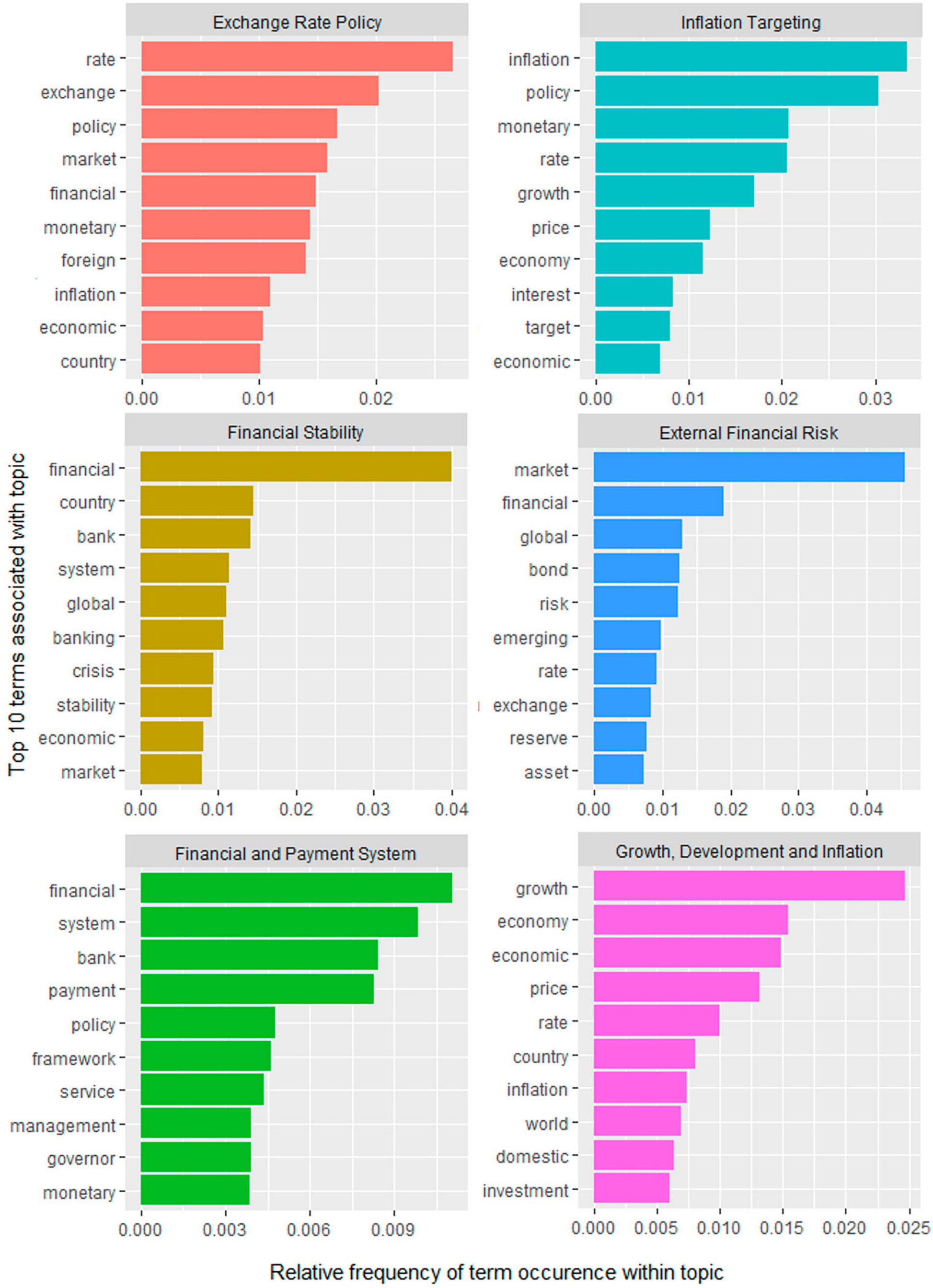


Figure 6. Topic analysis for speeches: top 10 terms associated with six topics.

The formal changes in the SARB's mandate (represented by dotted, vertical lines in Figure 5(a), 5(b)), also cannot easily be used to explain sudden changes in readability. But it is quite reasonable to argue that the introduction of the financial stability mandate, which was preceded by the 2007–2009 global financial crisis, increased the complexity of language for a while thereafter. For example, in March 2009 the SARB announced that they would hold their regular MPC meetings more often so as 'to allow for more effective planning and quicker reactions to changes in the economy' (MPC statement March, 2009). Then, in September 2010, it was becoming more evident that policy rates would 'remain abnormally low for an extended period of time in a number of the advanced economies' and, somewhat uncharacteristically, on 18 November 2010 Governor Marcus explicitly addressed the heated and wide-ranging public discourse on the appropriate monetary policy response:

> In considering the prevailing conditions and data, it is important to emphasise that the focus, notwithstanding any immediate impact or influence, is on the situation that might exist 12–18 months hence. There are significant voices and varied opinions aired in the public domain, all expressing a view on what needs to be done. While it is important to listen to and hear what is being said, it is the responsibility of the Monetary Policy Committee to determine the path of interest rates without fear or favour. And this is what we will continue to do.

With the support of results from the topic analysis presented in the following section, we argue that the three periods in which the complexity increases are periods following major crises. In each of these cases, it was not business as usual and the SARB relied on more technical language to clarify its role and its application of monetary policy with respect to the exchange rate, financial stability, and the Covid-19 pandemic.

Scope

In this section we present analyses of the content of speeches and MPC statements of the SARB employing a standard method of statistical topic analysis based on generative probabilistic models for corpora of documents, known as latent Dirichlet allocation (LDA). This is an unsupervised machine learning algorithm that statistically identifies a fixed set of topics within a corpus. This method assumes that each document in a corpus is composed of probabilistic draw from a set of topics, where each topic is composed of a probability distribution of words. The algorithm then simultaneously assigns a probability that a given word is part of a specific topic, and determines how much of a document is composed of each of the identified topics, assuming topics are independently distributed within and across documents (Blei, Ng, and Jordan 2003).

The speeches and MPC statements published on the SARB website are full texts, which means they contain articles, prepositions etc. that signify no identifying content related to issues of concern to the bank. These standard English *stop words* need to be removed in an abstract statistical analysis of policy content. Moreover, the purpose of topic analysis in this context is to identify distinct topics (typically the policy foci of the bank) and to evaluate how the focus of the speeches and MPCs evolve over time. For this reason, many other words need to be removed that do not directly relate to the topics of interest and confound the identification of important, policy-related words (eg qualifiers such as 'upward' or names of governors).[24] Additionally, all plurals are cast into their singular form.

The two corpora of texts, speeches and MPSs, have quite different features. After preprocessing, the corpus of MPC statements contain only 4806 unique words, whereas the

[24]See online appendix A for the full list of removed words.

corpus of speeches contains 22,566. This suggests a quite reasonable presumption that is borne out by all the analyses of this section: the MPC statements are carefully focused policy documents that tend to use more similar language and focus on specific policy related topics that are consistent throughout the sample, whereas the speeches deal with a wider array of issues in more distinctive styles that vary discernibly across the sample (moreover, they are likely written by a wider variety of writers).

Topic analysis

When performing a statistical topic analysis via LDA of a corpus of documents, a parameter that must be chosen *ex ante* by the researcher is the number of topics that the algorithm should assign words to. After some experimentation, we follow Coco and Viegi (2020) in using six topics. Our rationale for this choice is twofold: (1) whether a concise historical narrative can be read from the topics identified by the (unsupervised) algorithm, and (2) whether the narratives potentially obtained from the MPC statements can be sensibly related to that obtained from the speeches. Our results suggest that this agnostic, statistical approach is informative for the speeches released by the SARB but not for the MPC statements, which puts us at odds with the results in Coco and Viegi (2020).

Topic analysis of the MPC statements

Statistical topic analysis of the MPC statements is a very different exercise from that of the corpus of speeches. The corpus of speeches is diverse in terms of speaker, topic, and audience, whereas the MPC statements are systematic documents focused on the dissemination and motivation of policies aimed at the price stability mandate of the SARB. Indeed, the sample of MPC statements corresponds almost entirely with the period of formal inflation targeting, as these statements were designed as the primary tool of the SARB for communicating and motivating their decisions on the changes to the repo rate.

Applying the LDA algorithm to the corpus of MPC statements yielded six topics that were indistinguishable from one another with any degree of confidence. Words like 'monetary' 'inflation', 'growth', 'price', and 'rate' were in the top 10 terms of most of the six topics. Even after filtering out terms that are common to three or more of the top 60 terms in each of the statistically identified topics, the remaining words unique to each topic were no more informative (in the sense that they represent an obvious set of distinct topics). Rather, the terms identified seem to be a random selection of cost drivers from which we could not derive a clear narrative that relates to the economic history of South Africa over this period. We therefore relegate these results to online Appendix A.

This result is not surprising given the purpose and structure of the MPC statements, which is to motivate the policy decisions of the MPC in the face of all the potential sources of inflation. It is therefore quite natural that each MPC statement should cover the same set of topics – ie the standard potential drivers of future inflation. The variation to be expected within the corpus of MPC statements is not in *which* inflation drivers are discussed, but the current and forecast *state* of each inflation driver, which motivate the policy decision presented in that statement. Thus, while topic analysis of the MPC statements are unlikely to be illuminating, sentiment analysis of these documents is very valuable, as that method can indicate whether the prospects for inflation, for example, is positive or negative (as we show in the next section).

Topic analysis of speeches

Topic analysis of speeches yields more interesting results. The top terms associated with each topic can be relatively clearly interpreted as different parts of the various policy areas that affects the choices of the SARB in its pursuit of its mandates. To focus the algorithm on the most important aspects of the corpus, we trim the terms in the analysis to contain only those terms that occur in at least 50% of the documents in the corpus.[25] This reduces the total amount of terms in the database from 22,552 to 13,697. LDA is then employed to assign these terms to six topics. The top 10 terms (those with the highest probability to be from each topic) are presented in Figure 6.[26]

By inspection of the top 10 terms in each topic presented in Figure 6, we classify the core content of the topics identified as follows: [27]

Topic 1: Exchange rate policy	Topic 4: Inflation targeting
Topic 2: Financial Stability	Topic 5: External financial risk
Topic 3: Financial and payment system	Topic 6: Growth, development and inflation

How important are these various issues over time? We study this by estimating the probability weight of each topic in each speech and then averaging over the speeches in each year. This yields Figure 7.

These developments correspond quite closely to what one might expect in the historical context of South Africa. Issues related to exchange rate policy (topic 1) dominated in the early part of the sample, mirroring the SARB's active management of the exchange rate and the exchange rate crises in the 1990s and early 2000s, while the relative importance of this topic waned over time as the financial system recovered and became robust and internationally integrated. This also corresponds to the period before inflation targeting where monetary policy followed a more eclectic approach.

Financial stability (topic 2) and external financial risk (topic 5) are clearly relatively important around times of crises (the rand crises of 1996 and 2001, the global financial crisis from 2007 to 2009, the multi-year European sovereign debt crisis from 2010, the taper tantrum of 2013, and the collapse of African Bank in 2014). More generally, these topics are also persistently more important after the adoption of the financial stability mandate of the SARB in 2010.

Inflation targeting (topic 3) gains prominence over time from the first MPC meeting (1999), with a noticeably dominant position in 2020. By manual examination of the speeches, the reason for the latter appears to be strongly related to mounting public and political pressure to change the mandate of the SARB (eg to include targeting real economic outcomes such as employment) and to increase quantitative easing in response to the Covid-19 crisis.[28] There was also a concerted effort to convey inflation targeting as

[25]A standard approach in this method is to reduce the size of the corpus before analysis. This speeds up the algorithm considerably, and avoids confounding the topic identification by many terms that are infrequently used.

[26]Since we use the entire sample to identify this small number of topics, they should be interpreted as broad themes that are dominant over a large part of the period. Brief events that are important for only a short time (eg the failure of African Bank in 2014) will not be picked up as distinct topics.

[27]Note that the ordering of the statistically generated topics is arbitrary and do not indicate the relative importance of the topics in the corpus.

[28]For example, a speech on 18 June 2020 by Lesetja Kganyago, titled 'The coronavirus shock, and "the age of magic money"' discusses criticism in the media on the SARB's policy response and mounting political pressure to increase quantitative easing; and on 12 August 2020 a speech by Governor Kganyago, titled 'In the shadow of COVID:

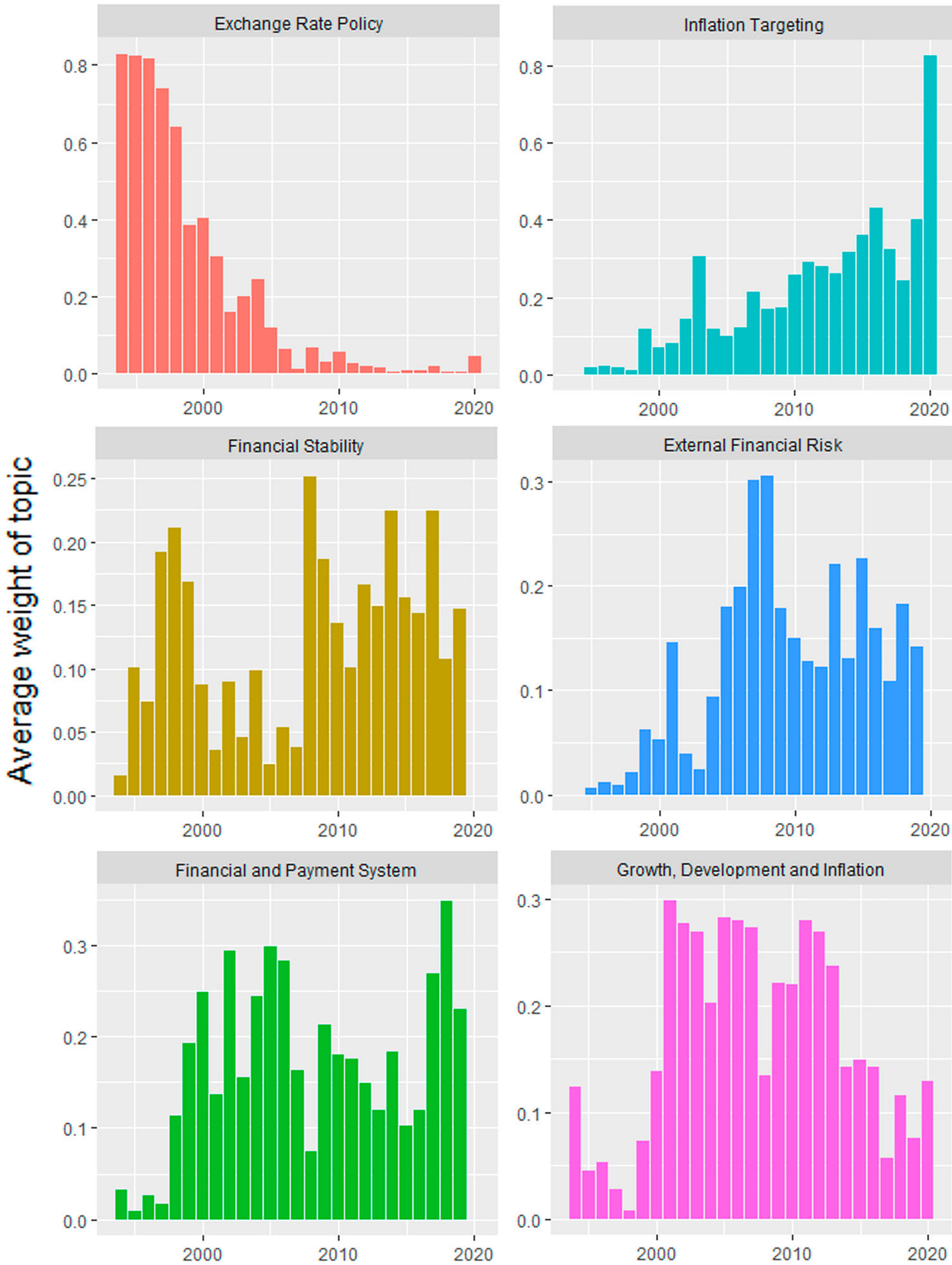


Figure 7. Topic analysis for speeches: evolution of topic frequency in speeches over time.

an 'expression of the mandate' and to contextualize the rationale behind targeting the mid-point of the 3–6% target band (SARB speech, 6 March 2019). That is, in the above context and given its mandate, SARB policymakers are likely to emphasize and strengthen

Lessons from 20 years of inflation targeting', discusses public pressure to change the mandate and implement quantitative easing.

its successes with inflation targeting. These issues were discussed throughout 2019 until 30 November 2020.[29]

While we labelled topic 3 'Financial and Payment system', it also contains terms related to monetary policy. On the payment system side of the interpretation, it shows a clear spike in 2017/2018 which corresponds to the time of Financial Sector Regulation Act in 2017 and the implementation of the strategic plan for the national payment system in 2018 (South African Reserve Bank 2018). Interpreting the financial and payment system as an important transmission channel of monetary policy, and taking it in combination with the other inflation topics: inflation targeting (topic 3) and the interaction between growth, development and inflation (topic 6) clearly shows the dominant nature of the core price stability mandate of the SARB reflected in the speeches from 1999 to the present.

Our results from employing statistical topic analysis thus shows it is useful in analysing the more eclectic coverage of the speeches released by the SARB, and it is less useful for the formal and more consistently structured MPC statements. We therefore turn to sentiment analysis which measures the tone of language (positive/negative) to extract information on the views of policymakers and how this relates to active policy choices.

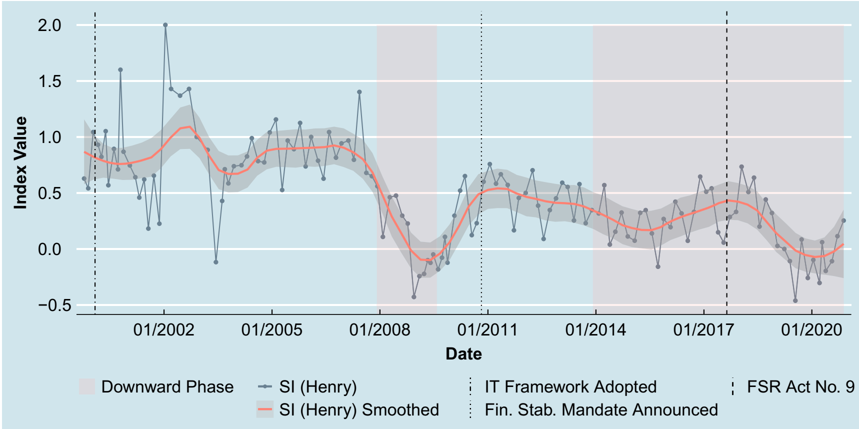
Sentiment

The SARB's Monetary Policy Committee was established in 1999 for the formulation of monetary policy and, in doing so, the MPC was tasked with communicating the stance of monetary policy (SARB MPC October, 1999). As such, text-mining techniques can be used to gauge the 'sentiment' of MPC statements and therefore provide information on the current and future stance of monetary policy (ie policy rate decisions). This approach was studied in Erasmus and Hollander (2020), and we refer the reader to this source for a comprehensive review of the literature, as well as the central bank policy of forward guidance.[30] Here, we compare the sentiment indices from MPC statements and speeches, as well as both corpus' utilization of the library ('bag of words') that the index is derived from. A priori, we would expect MPC statements to reflect the stance of monetary policy more accurately than speeches, since speeches cover a wider range of issues. Indeed, MPC statements should reflect the state of the economy and be predominantly driven by the current and expected inflation dynamics over the medium term (Erasmus and Hollander 2020).

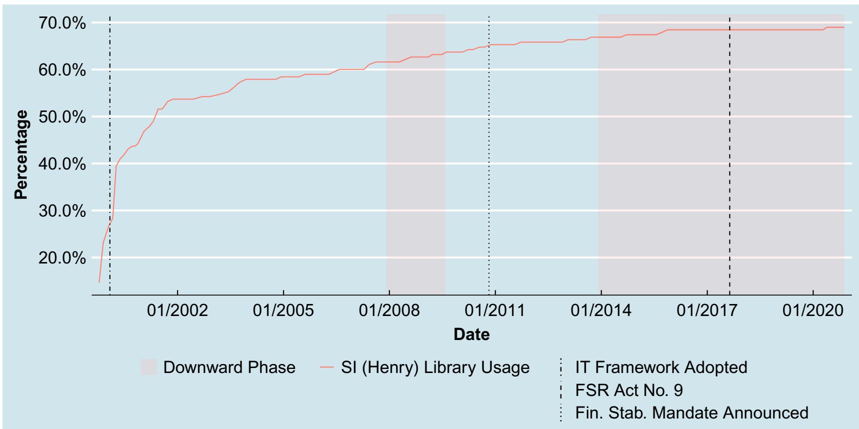
Figure 8(a) plots the derived sentiment index for MPC statements, where more positive (negative) values indicate a more 'hawkish' ('dovish') stance. It is evident that the

[29]For example, on 6 March 2019 Governor Kganyago spoke on 'Why central bank independence matters for that broader public good [monetary policy] and how it relates to achieving our inflation-targeting objective'. On 26 July 2019, the 'nationalization' of the SARB is discussed at the AGM. On 30 November 2020, Dr Rashad Cassim (Deputy Governor of the SARB) delivered a keynote address at the 3rd Annual HSBC Africa Conference on 'Lessons from the crisis and the post-COVID-19 outlook for monetary policy'.

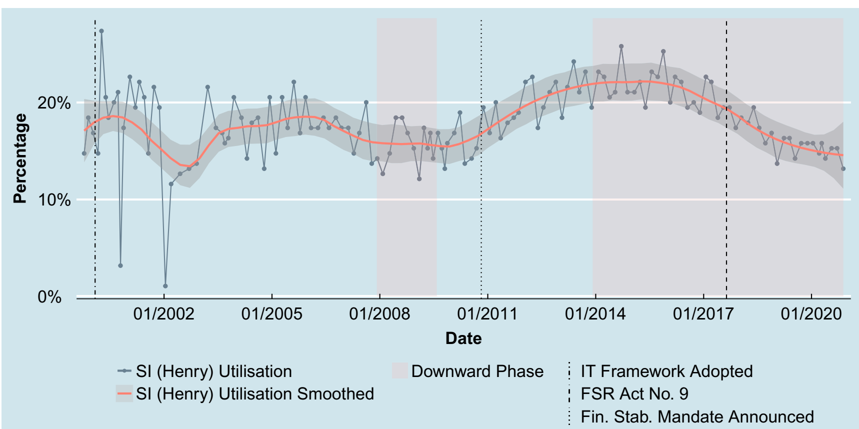
[30]Reid and Plessis (2010) use manual coding to create an 'inclination index', capturing the communicated inclination of future policy in the MPC statements. Given that in this application to monetary policy, the sentiment index of Erasmus and Hollander (2020) is in fact being used to measure the likely future path of policy, the two are in fact aiming to achieve the same thing. The results of the text mining approach in Erasmus and Hollander (2020) follow a similar pattern to those of Reid and Plessis (2010) where their sample overlap, giving greater confidence in the validity of the text mining results.



(a)



(b)



(c)

Figure 8. Sentiment analysis for MPC statements. (a) SI for MPC (with economic cycles). (b) Library evolution for MPC (with economic cycles). (c) Library utilization MPC (with economic cycles).

Notes: The special meeting held in January 2002 contains only a single paragraph, in which only two words are matched with the library, which results in a near-zero utilization of the library and a SI-score of 2. The April 2000 MPC meeting contains a detailed appendix discussion on the new inflation targeting framework. Library evolution refers to the cumulative share of library words that have been used up to a given point.

sentiment index (SI) falls during downward economic phases.[31] That is, the implied monetary policy stance is more accommodative (less hawkish; more dovish). This is in accordance to what we would expect since the SARB would likely reduce rates during times of economic downturn to stimulate economic activity. We can also infer from the figure that MPC statements have an inherent anti-inflation stance – evident from the SI value being predominantly above zero. However, this particular inference requires additional analysis given that the SI value is dependent on both the content of the speech and the composition of the sentiment library (Erasmus and Hollander 2020). Also, we notice that the SI becomes significantly more accommodative during the global financial crisis (GFC), from which, after the announcement of the financial stability mandate, the level of the index remained significantly lower than the pre-GFC period. The lower level of MPC sentiment (ie a more dovish stance) is very likely a function of (1) the relative importance of output and employment as a topic since 2014 (Figure 7), (2) the inclusion of financial stability in the SARB's mandate, and (3) the observed decline in both the volatility and level of inflation (Figure 3).[32] Finally, the variability of the SI for MPC statements is much more constrained than for speeches, which reflects the inherent nature of MPC statements: to convey a clear and consistent message (or sentiment) on the stance of monetary policy.

Figures 8(b) and 9(b) show the extent to which the corpora utilize the library of words used to derive the indices. As time progresses, both MPC statements and speeches encompass a richer variety of words, but a far greater proportion of the library is used by speeches (95% for speeches versus 65% for MPC statements). Interestingly, the cumulative usage of the library by the corpora flattens out and, by the end of 2020, it appears that they have exhausted the diversity of language necessary for effective communication. This is somewhat substantiated by the fact that clarity of communication trumps the volume of communication.

At the same time, only a fraction of the library is used for a given MPC statement or speech (see Figures 8(c) and 9(c)).[33] MPC statements typically range between 10 and 20% utilization, whereas speeches range anywhere between 0 and 40%. This reflects the fact that speeches tend to be longer than MPC statements (see Figure 4(b)).[34] Interestingly, we see the utilization rate for speeches increase after the adoption of the inflation targeting mandate, which suggests a richer communication as a result of the emphasis the inflation targeting framework places on transparency and communication.[35] For MPC statements, however, the library utilization remained fairly stable. This stability is likely because of its consistent structure and that communication about the inflation targeting framework had been in effect for

[31]The rise and fall of the SI from January 2002 reflects the concerns over rand depreciation pressures. Economic growth also fell from 5.3% in January 2002 to 2% in February 2003 (annualized quarter-on-quarter GDP growth). The special meeting held in January 2002 contains only a single paragraph, in which only two words are matched with the library, which results in a near-zero utilization of the library and a SI-score of 2. This 'hawkish' stance, although unrealistically high, does substantiate the 1 percentage point increase in the policy rate at the meeting.

[32]We can infer that governor tenures likely do not drive sentiment in either speeches or MPC statements, but rather the economic climate (see Figures S9(a) and S8(b)). That said, the tenure changes have typically followed very soon after economic downswings. A more sophisticated analysis should be done. We leave this for future research.

[33]As pointed out by Erasmus and Hollander (2020) in relation to the choice of library, this finding further substantiates the need for a library that is more tailored to the South African context.

[34]We do see that utilization might be somewhat dependent on Governor tenure, something we leave for future econometric analysis (see Figure S8(a)).

[35]It is very likely that, coming off a low utilization rate base, the marginal benefit of using an additional word in the library is high; in fact, the Flesch-Kincaid Grade score fell during this period (see Figure 5(b)) and the number of speeches was relatively low (see Figure 4(a)).

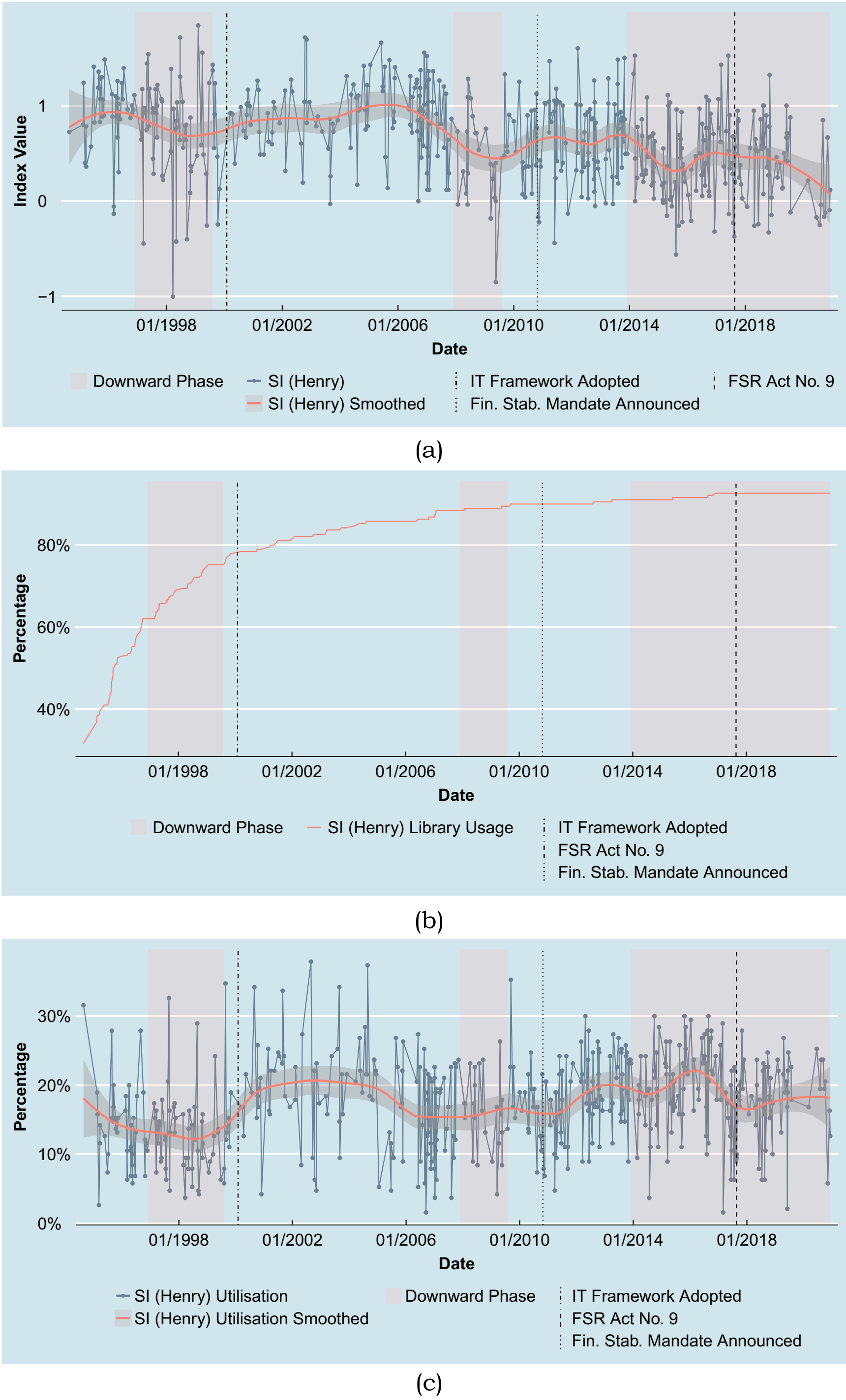


Figure 9. Sentiment analysis for speeches. (a) Sentiment index (with economic cycles). (b) Library evolution (with economic cycles). (c) Library utilization (with economic cycles).

Notes: Library evolution refers to the cumulative share of library words that have been used up to a given point.

nearly a decade. In contrast, after the GFC and the announcement of the new extended mandate, library utilization went up considerably before peaking at the onset of the most recent downswing. By the time of the FSR Act No 9, the utilization rate of the library by MPC statements had returned to its trend rate from 20 to 15%.

Concluding remarks

We relate our findings within the overarching context of increased transparency and communication (the 'first wave') and the extent to which the SARB considers the general public's reception of its communication (the 'second wave'). In addition to discussing the quantity and clarity of central bank (CB) communication and the recognition of its audience, we discuss how the state of the economy (eg the distress brought about by the global financial crisis) influences CB communication and how the SARB uses its CB communication to reinforce the status quo of its institutional design and operational procedures.[36]

Results from a range of data mining techniques allow us to characterize the SARB's communication along a number of dimensions: the quantity of documents and words (volume), the accessibility of the communication (linguistic complexity), the scope of the communication content (topics), and the extent to which the communication provides insight into the stance of policy (sentiment). The MPC statements differ notably from the speeches, in ways that are generally consistent with their more narrow purpose. In contrast, there is a lot more variability in speeches.

We find that the frequency of MPC statements remains very stable over time: typically six are scheduled per year, with additional meetings only being held during periods of economic distress or to clarify institutional changes that affect the implementation of policy. In all of these cases, this communication seeks to reduce uncertainty in markets or with the public. The word count of MPC statements appears to also vary with the state of the economy or with the introduction of institutional changes. Notably, after the global financial crisis and the announcement of the financial stability mandate, the total word count rose by a third from around 2000 to 3000 words per document. The linguistic complexity (readability) of the MPC statements also varies over time, peaking after the exchange rate crisis and the global financial crisis, which is not surprising given that new technical concepts would have needed to be communicated to contextualize and explain the policy responses. The scope (content) of the MPC statements is very stable over time, not allowing us to identify clearly distinctive topics using topic analysis in this study. Network plots reveal that the statements are strongly focused on the mandate of the SARB, indicating a consistent focus in the statements on the SARB's mandate. As expected, MPC statements reflect the policy stance of the SARB more clearly than speeches. The library used for the sentiment analysis also reveals that the MPC statements use a smaller proportion of the words in the library, again reflecting their more focused nature.

In contrast, our results show that the level and variability of the volume of speeches is much higher than in the case of the MPC statements, with the frequency of speeches increasing particularly after 2003. The word count of these documents also tends to be higher, and they contain language that is generally more complex. This higher quantity

[36]For broad historical discussions on South Africa's institutional design and operational procedures for financial stability, see eg Gilbert, Calitz, and du Plessis (2009); Hollander and van Lill (2019).

and limited accessibility is very likely because these speeches are directed at more specialized audiences in academia, at policy and society conferences, and the AGM, for example. We also observe that the linguistic complexity of the speeches increases in the periods following the three major crises in the samples. This likely reflects that speeches were one of the tools used to clarify the SARB's role and its application of monetary policy with respect to the exchange rate, financial stability, and the Covid-19 pandemic. These speeches are likely to have been more technical in nature. The network plot of the speeches suggests that speeches cover a more dispersed set of topics with no obvious dominant focus. Using a machine learning algorithm, we then identify six distinguishable topics; our analysis again suggests that speeches were used during times of distress to address the pressure being experienced by the economy at the time. Finally, it is difficult to interpret the sentiment index for speeches because of their variety. This broader focus is substantiated by the fact that the proportion of words matched with the dictionary is far higher.

Reflecting on the collection of findings and their relation to the historical context (academic, institutional, and economic events), we conclude that MPC statements have been narrowly focused on the mandate of the SARB and do so in a very consistent manner. There is evidence that this communication adjusts in response to crises and relevant institutional changes such as the expansion of the mandate to include financial stability, but this is done in a very prudent and predictable manner.

The speeches capture more detail about how the thinking of SARB policymakers is evolving over time. The topics addressed in these speeches are still focused on topics relevant to monetary policy, but they address more deliberately the pressures of the time. We can see the discussion of inflation targeting in speeches built over time, rather than dominating from the beginning. This may have been a deliberate decision to speak with one voice (to avoid the cacophony problem [Blinder 2004]) while the inflation targeting framework was becoming more established and it may also reflect learning-by-doing. The latter is supported by the evolving structure of the MPC statements. We also see that although financial stability becomes an important focus after the global financial crisis, the topic was discussed in speeches with only slightly less intensity in response to the Rand crises of 1996 and 2001. This reflects that the SARB was never blind to the importance of financial stability. The cognisance of the SARB to the need for economic growth in South Africa, and its role in providing a stable background using a flexible inflation targeting regime to enable this, is also clear.

Disclosure statement

No potential conflict of interest was reported by the authors.

ORCID

Hylton Hollander http://orcid.org/0000-0002-5555-9050

References

Bernanke, B. S., T. Laubach, F. S. Mishkin, and A. S. Posen. 1999. *Inflation Targeting: Lessons from the International Experience*. Princeton: Princeton University Press.

Bholat, D., S. Hans, P. Santos, and C. Schonhardt-Bailey. 2015. "Text Mining for Central Banks, Number 33." In *Handbooks*. Centre for Central Banking Studies, edited by Andrew Blake, 1–21. London: Bank of England.

Bhundia, A., and L. Ricci. 2006. "The Rand Crises of 1998 and 2001: What Have We Learned?" In *Post-Apartheid South Africa: The First Ten Years*, edited by M. Nowak and L. Ricci, 156–173. Washington: International Monetary Fund.

Binder, C. 2017. "Fed Speak on Main Street: Central Bank Communication and Household Expectations." *Journal of Macroeconomics* 52: 238–251.

Blei, D. M., A. Y. Ng, and M. I. Jordan. 2003. "Latent Dirichlet Allocation." *Journal of Machine Learning Research* 3 (January): 993–1022.

Blinder, A. S. 2004. *The Quiet Revolution: Central Banking Goes Modern*. New Haven, CT: Yale University Press.

Blinder, A. S. 2018. "Through a Crystal Ball Darkly: The Future of Monetary Policy Communication." *AEA Papers and Proceedings* 108: 567–571.

Blinder, A. S., M. Ehrmann, M. Fratzscher, J. De Haan, and D.-J. Jansen. 2008. "Central Bank Communication and Monetary Policy: A Survey of Theory and Evidence." *Journal of Economic Literature* 46 (4): 910–945.

Bordo, M. D., and P. Siklos. 2014. "Central Bank Credibility, Reputation and Inflation Targeting in Historical Perspective." National Bureau of Economic Research Working Papers No. 20693. http://www.nber.org/papers/w20693.

Bordo, M. D., and P. Siklos. 2019. "The Transformation and Performance of Emerging Market Economies Across the Great Divide of the Global Financial Crisis." National Bureau of Economic Research Working Papers No. 26342. http://www.nber.org/papers/w26342.

Borio, C. 2019a. "Central Banking in Challenging Times, Speech." https://www.bis.org/speeches/sp191108a.pdf.

Borio, C. 2019b. "Monetary Policy Frameworks in EMEs: Practice Ahead of Theory, Speech." https://www.bis.org/speeches/sp190630a.htm.

Bulíř, A., M. Cĭhák, and D. Jansen. 2013. "What Drives Clarity of Central Bank Communication About Inflation?" *Open Economies Review* 24 (1): 125–145.

Clarida, R., J. Galí, and M. Gertler. 1999. "The Science of Monetary Policy: A New Keynesian Perspective." *Journal of Economic Literature* 37 (4): 1661–1707.

Coco, A., and N. Viegi. 2020. "The Monetary Policy of the South African Reserve Bank: Stance, Communication and Credibility." South African Reserve Bank Working Paper Series No. WP/20/06. https://https://www.resbank.co.za/.

Coibion, O., Y. Gorodnichenko, and R. Kamdar. 2018. "The Formation of Expectations, Inflation, and the Phillips Curve." *Journal of Economic Literature* 56 (4): 1447–1491.

Cukierman, A., and S. B. Webb. 1995. "Political Influence on the Central Bank: International Evidence." *The World Bank Economic Review* 9 (3): 397–423.

de Haan, J., and S. Eijffinger. 2016. "The Politics of Central Bank Independence." De Nederlandsche Bank Working Papers No. 539. https://ideas.repec.org/p/dnb/dnbwpp/539.html.

Dincer, N. N., and B. Eichengreen. 2014. "Central Bank Transparency and Independence: Updates and New Measures." *International Journal of Central Banking* 10 (1): 189–259.

Draghi, M. 2019. "Code of Conduct for High-Level European Central Bank Officials (2019/C 89/03)." *Official Journal of the European Union C* 89 (2). https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52019XB0308(01).

Eichengreen, B. 2012. "Economic History and Economic Policy." *The Journal of Economic History* 72 (2): 289–307.

Erasmus, R. and H. Hollander. 2020. "A forward guidance indicator for the South African Reserve Bank: implementing a text analysis algorithm." *Studies in Economics and Econometrics* 44 (3): 41–72. DOI: 10.1080/03796205.2020.1919424.

Flesch, R. 1948. "A New Readability Yardstick." *Journal of Applied Psychology* 32 (3): 221–233.

Geraats, P. M. 2002. "Central Bank Transparency." *The Economic Journal* 112 (483): F532–F565.

Gilbert, E., E. Calitz, and S. du Plessis. 2009. "Prudential Regulation, Its International Back-Ground and the Performance of the Banks: A Critical Review of the South African Environment Since 1970." *South African Journal of Economic History* 24 (2): 43–81.

Graham, S., I. Milligan, and S. Weingart. 2016. *Exploring Big Historical Data: The Historian's Macroscope*. London: Imperial College Press.

Gürkaynack, R., A. Levin, A. Marder, and E. Swanson. 2007. "Inflation Targeting and the Anchoring of Inflation Expectations in the Western Hemisphere." *Federal Reserve Bank of San Francisco Economic Review* 2007: 25–47.
Gürkaynack, R., B. Sack, and E. Swanson. 2005a. "The Sensitivity of Long-Term Interest Rates to Economic News: Evidence and Implications for Macroeconomic Models." *American Economic Review* 95 (1): 425–436.
Gürkaynack, R., B. Sack, and E. Swanson. 2005b. "Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements." *International Journal of Central Banking* 2005 (May): 55–93.
Gürkaynak, R. S., A. Levin, and E. Swanson. 2010. "Does Inflation Targeting Anchor Long-Run Inflation Expectations? Evidence from the U.S., UK, and Sweden." *Journal of the European Economic Association* 8 (6): 1208–1242.
Haldane, A. 2017. "A Little More Conversation, A Little Less Action." Speech Given at the Federal Reserve Bank of San Francisco Macroeconomic and Monetary Policy Conference on 31 March.
Haldane, A., A. Macaulay, and M. McMahon. 2020. "The 3E's of Central Bank Communication With the Public." Bank of England Working Papers No. 847. https://www.bankofengland.co.uk/.
Haldane, A., and M. McMahon. 2018. "Central Bank Communications and the General Public." *AEA Papers and Proceedings*, 108: 578–583.
Hammond, G. 2012. "State of the Art of Inflation Targeting, Number 29." In *Handbooks*, edited by Andrew Blake and Francesco Zanetti, 1–45. London: Centre for Central Banking Studies, Bank of England.
Henry, E. 2008. "Are Investors Influenced By How Earnings Press Releases Are Written?" *Journal of Business Communication* 45 (4): 363–407.
Hollander, H., and D. van Lill. 2019. *A Review of the South African Reserve Bank's Financial Stability Policies*. Sandton: Juta & Co and the Centre for Banking Law.
International Monetary Fund. 1999. "Code of Good Practices on Transparency in Monetary and Financial Policies: Declaration of Principles, Standards and Codes." https://www.imf.org/external/np/mae/mft/code/index.htm.
International Monetary Fund. 2020. "The Central Bank Transparency Code." Policy Paper No. 2020/038. https://www.imf.org/en/Publications/Policy-Papers/Issues/2020/07/29/The-Central-Bank-Transparency-Code-49619.
Issing, O. 2019. *The Long Journey of Central Bank Communication*. Cambridge: MIT Press.
Kincaid, J., R. Braby, and J. Mears. 1988. "Electronic Authoring and Delivery of Technical Information." *Journal of Instructional Development* 11: 8–13.
King, M. 2005. "Monetary Policy – Practice Ahead of Theory, Speech." https://www.bis.org/review/r050520a.pdf.
Krippendorff, K. 2013. *Content Analysis: An Introduction to Its Methodology*. 3rd ed. Thousand Oaks: SAGE Publications.
Larrain, M. 2005. "Monetary Policy and Long-Term Interest Rates in Chile." Central Bank of Chile Working Paper No 335.
Mboweni, T. T. 2003. "Inflation Targeting in South Africa, Speech." https://www.bis.org/review/r030909e.pdf.
Mishkin, F. 2006. "Inflation Targeting: True Progress or Repackaging of an Old Idea?" Working Paper. Columbia Business School. https://www0.gsb.columbia.edu/mygsb/faculty/research/pubfiles/1999/05snb.pdf.
Palley, T. 2012. "The Economics of the Phillips Curve: Formation of Inflation Expectations Versus Incorporation of Inflation Expectations." *Structural Change and Economic Dynamics* 23 (3): 221–230.
Reid, M. 2009. "The Sensitivity of South African Inflation Expectations to Surprises." *South African Journal of Economics* 77 (3): 414–429.
Reid, M., and S. D. Plessis. 2010. "Loud and CLEAR? Can We Hear When the SARB Speaks?" *South African Journal of Economics* 78 (3): 269–286.
Reid, M., and P. Siklos. 2020. "Building Credibility and Influencing Expectations – The Evolution of Central Bank Communication." Working Papers 10144. https://ideas.repec.org/p/rbz/wpaper/10144.html.

Reid, M., P. Siklos, T. Guetterman, and S. Du Plessis. 2021. "The Role of Financial Journalists in the Expectations Channel of the Monetary Transmission Mechanism." *Research in International Business and Finance* 55: 101320.

Reinhart, C., and K. Rogoff. 2004. "The Modern History of Exchange Rate Arrangements: A Reinterpretation." *The Quarterly Journal of Economics* 119 (1): 1–48.

Romer, C. D., and D. H. Romer. 1989. "Does Monetary Policy Matter? A New Test in the Spirit of Friedman and Schwartz." *National Bureau of Economic Research Macroeconomics Annual* 4: 121–170.

Shiller, R. J. 2017. "Narrative Economics." *American Economic Review* 107 (4): 967–1004.

Siklos, P. L. 2003. "Assessing the Impact of Changes in Transparency and Accountability at the Bank of Canada." *Canadian Public Policy* 29: 279–299.

Siklos, P. L. 2017. *Central Banks Into the Breach*. Oxford: Oxford University Press.

Siklos, P. L., S. S. Amand, and J. Wajda. 2018. "The Evolving Scope and Content of Central Bank Speeches." Centre for International Governance Innovation, Working Paper No. 202.

South African Reserve Bank. 2018. "The National Payment System Framework and Strategy Vision 2025, Press Statement." https://www.resbank.co.za.

South African Reserve Bank. 2020. "Various Statements Issued and Speeches Published by the South African Reserve Bank, Textual Data." Accessed November 30, 2020. https://www.resbank.co.za/.

Tarullo, A. 2017. "Monetary Policy Without a Working Theory of Inflation." Hutchins Center Working Papers No. 33. https://www.brookings.edu.

Walsh, C. 2015. "Goals and Rules in Central Bank Design." *International Journal of Central Banking* 11 (S1): 295–351.

Wehrheim, L. 2019. "Economic History Goes Digital: Topic Modeling the Journal of Economic History." *Cliometrica, Journal of Historical Economics and Econometric History* 13 (1): 83–125.

Winkler, B. 2002. "On the Need for Effective Communication in Monetary Policy." *Ifo Studien* 48 (3): 401–427.

Woodford, M. 2003. *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton: Princeton University Press.

Yellen, J. 2015. "Testimony Before the U.S. House Banking Committee, Speech." https://www.federalreserve.gov/newsevents/testimony/yellen20150224a.htm.